

Math II Exam

04.09.2024

1. (25 points) Consider the function $f(x) = \frac{x}{\ln x}$

- (a) Determine the domain of $f(x)$.
(b) Determine the following limits:

$$\lim_{x \rightarrow 0^+} f(x), \quad \lim_{x \rightarrow 1^-} f(x), \quad \lim_{x \rightarrow 1^+} f(x), \quad \lim_{x \rightarrow \infty} f(x).$$

- (c) Compute the minima and maxima of $f(x)$ without the use of the second derivative.
(d) Sketch the graph of $f(x)$ by using the above information.

2. (20 points)

- (a) Calculate the integral:

$$\int_0^{\frac{1}{3}} 27\pi x^2 e^{3x} dx$$

- (b) Calculate the limit

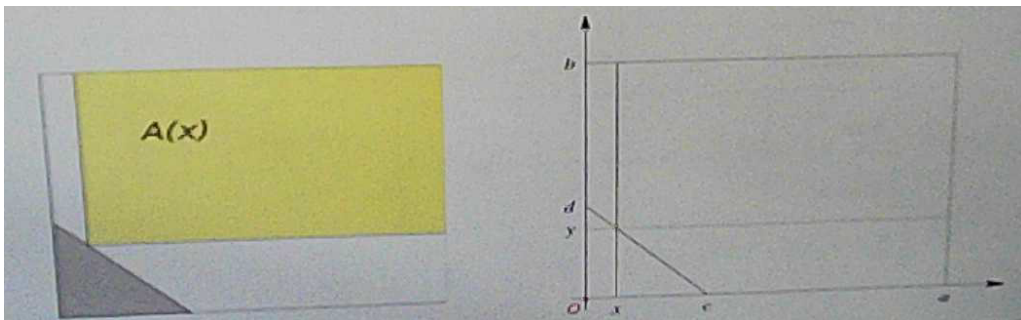
$$\lim_{x \rightarrow 0} \frac{e^{2x} - 2\sin(x) - 1 - 2x^2}{\sin(\pi x) - \pi x}$$

3. (10 points) Find all parameters $x \in \mathbb{R}$ such that the matrix

$$A = \begin{pmatrix} 1 & 2 & 0 \\ 2 & 5 & x-1 \\ 0 & x-1 & 1 \end{pmatrix}$$

is positive definite.

4. (20 points) The figure below depicts a rectangular glass plate where a triangular corner is broken off. The broken glass plate has width a and height b . The triangular corner has width c and height d such that $0 < c < a$ and $0 < d < b$. We would like to cut out of the remaining glass plate, a rectangular glass plate P such that this new plate has maximal area. The south-west corner O denotes the origin $(0, 0)$ of a coordinate system.



- (a) Express y as a function of x so that the graph of this function is the line that passes through $(0, d)$ and $(c, 0)$.
(b) Express the area of the rectangular glass plate P as a polynomial $A(x)$ in terms of x .
(c) Calculate $A'(x)$ and $A''(x)$.
(d) Determine $x^* \in \mathbb{R}$ such that the value of the polynomial $A(x^*)$ is as large as possible.
5. (25 points) For the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, find the global minimum and maximum on the closed and bounded set $\Omega = \{(x, y) : x^2 + y^2 = 4\}$, where

$$f(x, y) = \frac{1}{2}x^2y - 2y.$$